

Load Dataset

```
import pandas as pd
```

```
df = pd.read_excel("Dataset.xlsx")
```

```
print(df.head())
```

```
print(df.info())
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber
0	928 Main St	Debit Card	Shipped	TRK37947903
7				
1	823 Main St	Online	Shipped	TRK91186779
3				
2	512 Main St	Credit Card	Cancelled	TRK42903982
8				
3	275 Main St	Debit Card	Returned	TRK62788070
5				
4	668 Main St	Online	Delivered	TRK29241424
8				

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.10
1	SAVE10	Referral	302.70
2	FREESHIP	Email	2753.40
3	SAVE10	Facebook	273.19
4	SAVE10	Email	2504.04

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1200 entries, 0 to 1199
```

```
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	OrderID	1200 non-null	object
1	Date	1200 non-null	datetime64[ns]
2	CustomerID	1200 non-null	object
3	Product	1200 non-null	object
4	Quantity	1200 non-null	int64
5	UnitPrice	1200 non-null	float64
6	ShippingAddress	1200 non-null	object
7	PaymentMethod	1200 non-null	object
8	OrderStatus	1200 non-null	object

```
9   TrackingNumber    1200 non-null    object
10  ItemsInCart       1200 non-null    int64
11  CouponCode        891 non-null     object
12  ReferralSource    1200 non-null    object
13  TotalPrice        1200 non-null    float64
dtypes: datetime64[ns](1), float64(2), int64(2), object(9)
memory usage: 131.4+ KB
None
```

Checking Missing Values

```
print(df.isnull().sum())
```

```
OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
ItemsInCart      0
CouponCode       309
ReferralSource   0
TotalPrice       0
dtype: int64
```

Handling Missing Coupon Codes

```
df["CouponCode"] = df["CouponCode"].fillna("No Coupon")
```

```
print(df.isnull().sum())
```

```
OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
ItemsInCart      0
```

```
CouponCode      0
ReferralSource   0
TotalPrice       0
dtype: int64
```

Cheaking Duplicate Rows

```
print("Duplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 0
```

Verifying Unique Order IDs

```
print("Duplicate OrderIDs:", df["OrderID"].duplicated().sum())
```

```
Duplicate OrderIDs: 0
```

Standardizing Text Columns

```
text_cols = [
    "Product", "PaymentMethod", "OrderStatus",
    "CouponCode", "ReferralSource"
]

for col in text_cols:
    df[col] = df[col].astype(str).str.strip().str.title()
```

Format Date Column

```
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df["Date"] = df["Date"].dt.strftime("%Y-%m-%d")
```

Validate TotalPrice

```
df["CalculatedTotal"] = df["Quantity"] * df["UnitPrice"]
df["PriceMatch"] = df["CalculatedTotal"].round(2) ==
df["TotalPrice"].round(2)

print("Incorrect TotalPrice rows:", (~df["PriceMatch"]).sum())

Incorrect TotalPrice rows: 0
```

Removing Temporary Columns

```
df.drop(columns=["CalculatedTotal", "PriceMatch"], inplace=True)
```

Final Validation

```
print("Missing values:", df.isnull().sum().sum())  
print("Duplicate rows:", df.duplicated().sum())  
print("Duplicate OrderIDs:", df["OrderID"].duplicated().sum())
```

```
Missing values: 0  
Duplicate rows: 0  
Duplicate OrderIDs: 0
```

Saving Cleaned Dataset

```
df.to_csv("Cleaned_Dataset.csv", index=False)
```